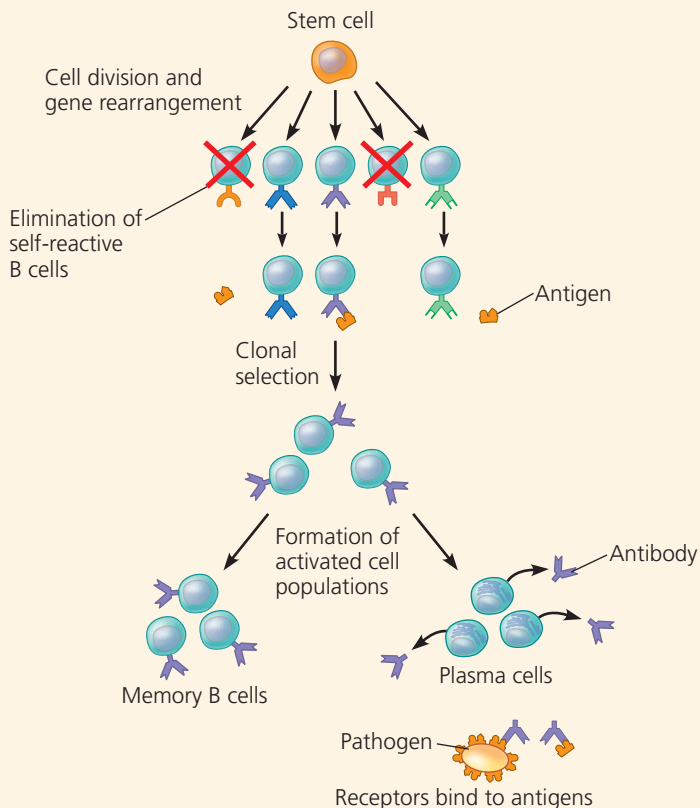
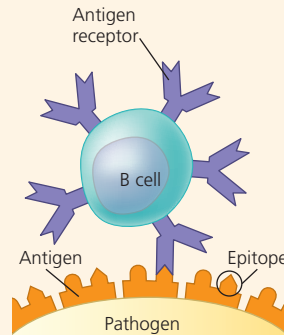


CONCEPT 43.2

In adaptive immunity, receptors provide pathogen-specific recognition (pp. 957–963)

- **Adaptive immunity** relies on two types of **lymphocytes: B cells and T cells**. Lymphocytes have cell-surface **antigen receptors** for foreign molecules (**antigens**). All receptor proteins on a single B or T cell are the same, but there are millions of B and T cells in the body that differ in the foreign molecules that their receptors recognize. Upon infection, B and T cells specific for the pathogen are activated. Some T cells help other lymphocytes; others kill infected host cells. B cells called **plasma cells** produce soluble proteins called **antibodies**, which bind to foreign molecules and cells. B and T **memory cells** defend against future infections by the same pathogen.
- Recognition of foreign molecules by B cells and T cells involves the binding of variable regions of receptors to an **epitope**, a small region of an antigen. B cells and antibodies recognize epitopes on the surface of antigens circulating in the blood or lymph. T cells recognize protein epitopes in small antigen fragments (peptides) that are presented on the surface of host cells by **major histocompatibility complex (MHC) molecules**. This interaction activates a T cell, enabling it to participate in adaptive immunity.
- The four major characteristics of B and T cell development are the generation of cell diversity, self-tolerance, proliferation, and immunological memory. Proliferation and memory are both based on **clonal selection**, illustrated here for B cells:



? Why is the adaptive immune response to an initial infection slower than the innate response?

CONCEPT 43.3

Adaptive immunity defends against infection of body fluids and body cells (pp. 963–970)

- **Helper T cells** interact with antigen fragments displayed by class II MHC molecules on the surface of **antigen-presenting cells**: dendritic cells, macrophages, and B cells. Activated helper T cells secrete cytokines that stimulate other lymphocytes. In the **humoral immune response**, antibodies help eliminate antigens by facilitating phagocytosis and complement-mediated lysis. In the **cell-mediated immune response**, activated **cytotoxic T cells** trigger destruction of infected cells.
- **Active immunity** develops in response to infection or to **immunization**. The transfer of antibodies in **passive immunity** provides immediate, short-term protection.
- Tissues or cells transferred from one person to another are subject to immune rejection. In tissue grafts and organ transplants, MHC molecules stimulate rejection. Lymphocytes in bone marrow transplants may cause a graft-versus-host reaction.

? Is immunological memory after a natural infection fundamentally different from immunological memory after vaccination? Explain.

CONCEPT 43.4

Disruptions in immune system function can elicit or exacerbate disease (pp. 970–974)

- In allergies, such as hay fever, the interaction of antibodies and allergens triggers immune cells to release histamine and other mediators that cause vascular changes and allergic symptoms. Loss of self-tolerance can lead to **autoimmune diseases**, such as multiple sclerosis. Inborn immunodeficiencies result from defects that interfere with innate, humoral, or cell-mediated defenses. **AIDS** is an acquired immunodeficiency caused by **HIV**.
- Antigenic variation, latency, and direct assault on the immune system allow some pathogens to thwart immune responses.

? Is being infected with HIV the same as having AIDS? Explain.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Which of these is absent from insect immunity?
 - antibacterial digestive enzymes
 - activation of natural killer cells
 - phagocytosis by hemocytes
 - production of antimicrobial peptides
- An epitope associates with which part of an antigen receptor or antibody?
 - the tail
 - the heavy-chain constant regions only
 - variable regions of a heavy chain and light chain combined
 - the light-chain constant regions only
- Which statement best describes the difference between responses of effector B cells (plasma cells) and those of cytotoxic T cells?
 - B cells confer active immunity; cytotoxic T cells confer passive immunity.
 - B cells respond the first time a pathogen is present; cytotoxic T cells respond subsequent times.
 - B cells secrete antibodies against a pathogen; cytotoxic T cells kill pathogen-infected host cells.
 - B cells carry out the cell-mediated response; cytotoxic T cells carry out the humoral response.